

Stat 4194 Project 1

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Data Selection

Name: Basketball Players Stats per Season - 49 Leagues

Sources: <https://www.kaggle.com/>, <https://basketball.realmgm.com/>

Citation: Baruch, J. (2020, November 24). Basketball players stats per season - 49 leagues. Kaggle. <https://www.kaggle.com/datasets/jacobbaruch/basketball-players-stats-per-season-49-leagues>

Website: <https://www.kaggle.com/datasets/jacobbaruch/basketball-players-stats-per-season-49-leagues>

Descriptive Abstract:

The “player_sub_by_season_full_details.csv” data set contains information regarding the different seasons from 1999-2020 categorized by 1 year blocks of 49 different Leagues, with the statistics of each season’s game such as field goals, 3 points, steals, blocks, points etc. The data set also includes 53,949 player information such as height, weight, nationality and their birth dates.

List of Variables: Description

League: League Name

Season: Season Year

Stage: International NBA, Playoffs & Regular Season

Player: Player Full Name

Team: Team Name

GP: Games Played

Min: Minutes Played

FGM: Field Goals Made

FGA: Field Goals Attempts

3PM: Three Points Made

3PA: Three Points Attempts

FTM: Free Throws Made

FTA: Free Throws Attempts

TOV: Turnovers

PF: Personal Fouls

ORB: Offensive Rebounds

DRB: Defensive Rebounds

REB: Rebounds
 AST: Assists
 STL: Steals
 BLK: Blocks
 PTS: Points
 birth_year: Birth Year
 birth_month: Birth Month
 birth_date: Birth Date
 height: Height [Feet]
 height_cm: Height [CM]
 weight: Weight [Pounds]
 weight_kg: Weight [KG]
 nationality: Nationality
 high_school: High School
 draft_round: Round
 draft_pick: Pick
 draft_team: Team selected in draft

Below is a small peak view of the data set for “player_sub_by_season_full_details.csv”:

```

players_stats <- read.csv("~/Python/Stats/STAT 4194/players_stats_by_season_full_details.csv")
player_sub <- players_stats[players_stats$Season=="2000 - 2001",]
head(player_sub, 4)
  
```

```

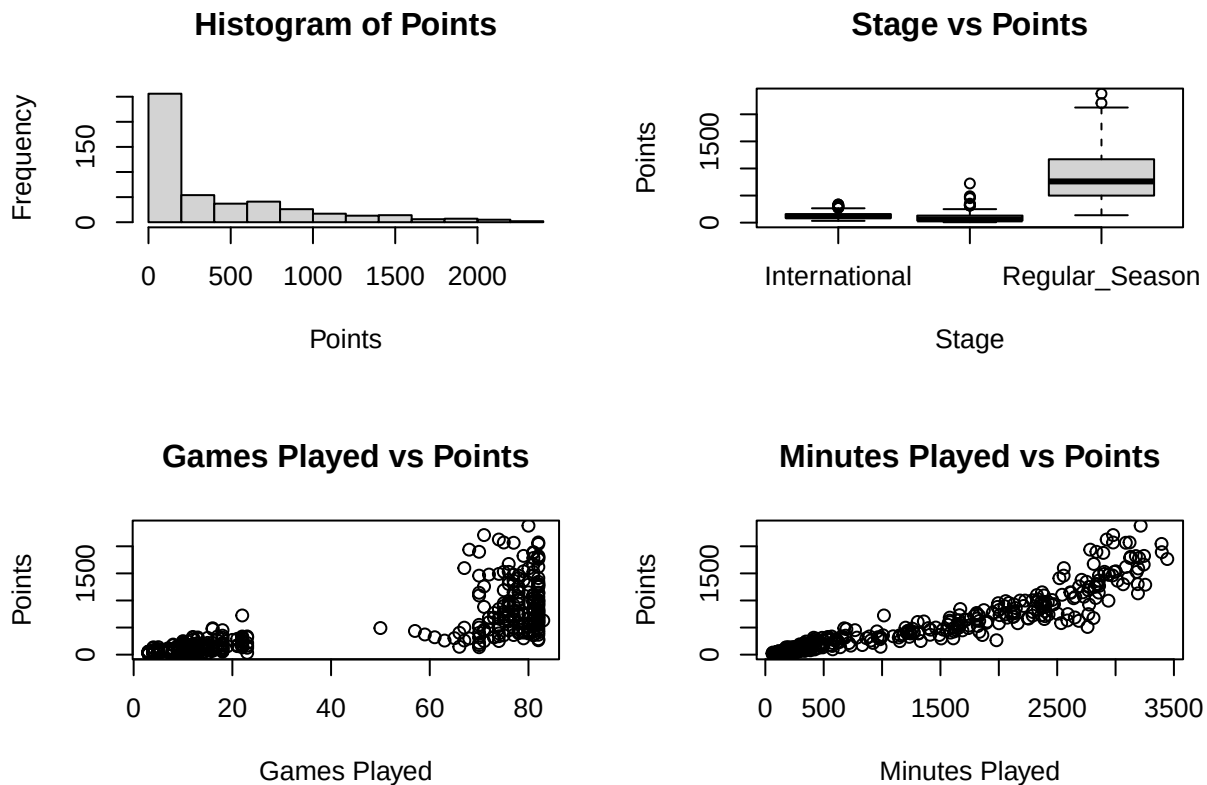
##           League      Season      Stage      Player Team GP   MIN FGM FGA
## 343 Euroleague 2000 - 2001 International Gregor Fucka  BOL 17 568.2 127 226
## 344 Euroleague 2000 - 2001 International Manu Ginobili  BOL 22 654.4 110 247
## 345 Euroleague 2000 - 2001 International Elmer Bennett  CAJ 22 784.3 100 254
## 346 Euroleague 2000 - 2001 International Alphonso Ford  PER 12 413.8 116 214
##           X3PM X3PA FTM FTA  TOV PF ORB DRB REB AST STL BLK PTS birth_year birth_month
## 343    12    31  69 110  38 37  42 101 143  12 39  14 335      1971      Aug
## 344    30   103  84 108  56 70  17  74  91  44 64   7 334      1977      Jul
## 345    22    84  92 124  56 58  12  37  49 120 43   1 314      1970      Feb
## 346    15    42  65  75  34 33  18  31  49  32 13   3 312      1971      Oct
##           birth_date height height_cm weight weight_kg      nationality
## 343 Aug 7, 1971    7-1      216      216      98 Italy / Slovenia
## 344 Jul 28, 1977    6-6      198      205      93      Argentina
## 345 Feb 13, 1970    6-0      183      171      78      United States
## 346 Oct 31, 1971    6-1      185      190      86      United States
##           high_school draft_round draft_pick      draft_team
## 343                NA          NA          NA
## 344                2           28      San Antonio Spurs
## 345      Bellaire High School      2           11      Atlanta Hawks
## 346 Amanda Elzy High School      2           5 Philadelphia Sixers
  
```

The data set was cut down to only 2000 - 2001 season. In this regression analysis, an exploration of the dependent outcome variable points (PTS) will be explained by different independent variables such as stage, field goals, blocks, steals, or player information such as height, weight, and draft picks.

Exploratory Analysis

Preliminary Variable Screening and Exploratory Plots:

```
#not normally distributed -- use glm, logistic reg. count(Poisson)
par(mfrow=c(2,2))
hist(player_sub$PTS, xlab="Points", main="Histogram of Points")
boxplot(PTS~Stage, data=player_sub,
        xlab="Stage", ylab="Points",
        main="Stage vs Points") #categorical predictor with 3+ levels
plot(x=player_sub$GP, y=player_sub$PTS,
     xlab="Games Played", ylab="Points", main="Games Played vs Points")
plot(x=player_sub$MIN, y=player_sub$PTS,
     xlab="Minutes Played", ylab="Points", main="Minutes Played vs Points")
```

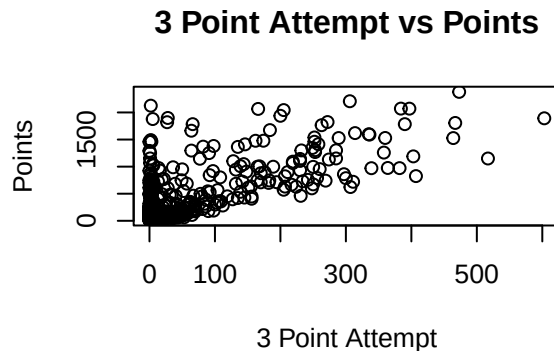
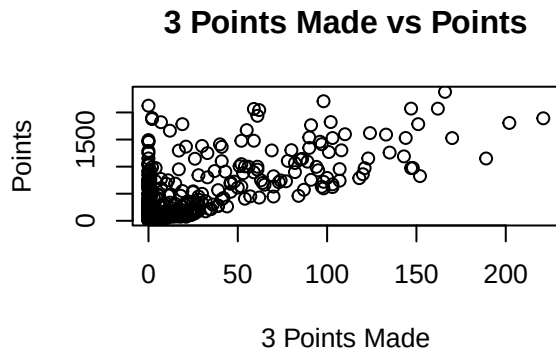
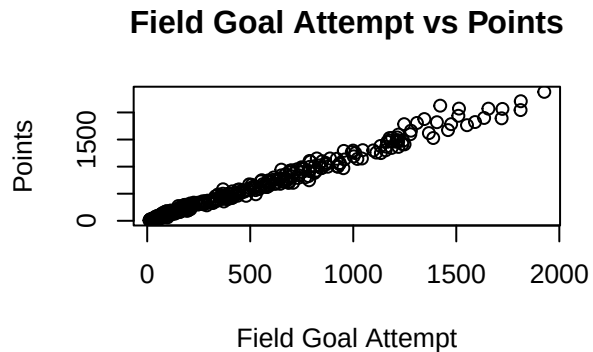
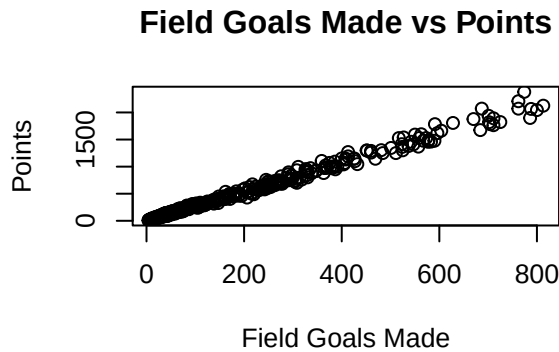


```
par(mfrow=c(2,2))
plot(x=player_sub$FGM, y=player_sub$PTS,
```

```

xlab="Field Goals Made", ylab="Points", main="Field Goals Made vs Points")
plot(x=player_sub$FGA, y=player_sub$PTS,
xlab="Field Goal Attempt", ylab="Points", main="Field Goal Attempt vs Points")
plot(x=player_sub$X3PM,y=player_sub$PTS,
xlab="3 Points Made", ylab="Points", main="3 Points Made vs Points")
plot(x=player_sub$X3PA,y=player_sub$PTS,
xlab="3 Point Attempt", ylab="Points", main="3 Point Attempt vs Points")

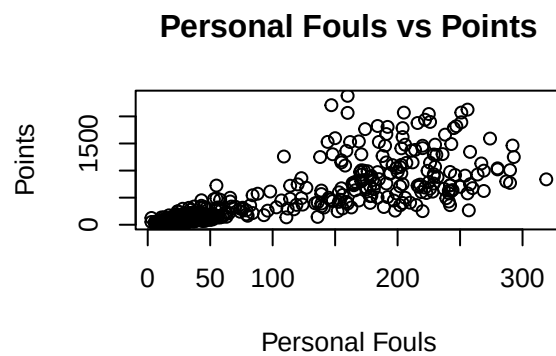
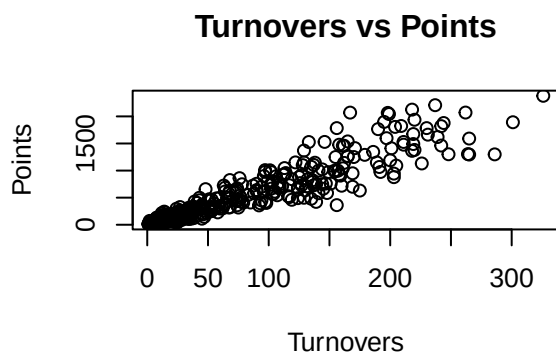
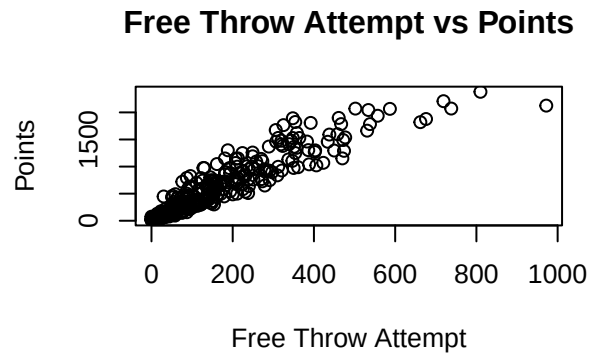
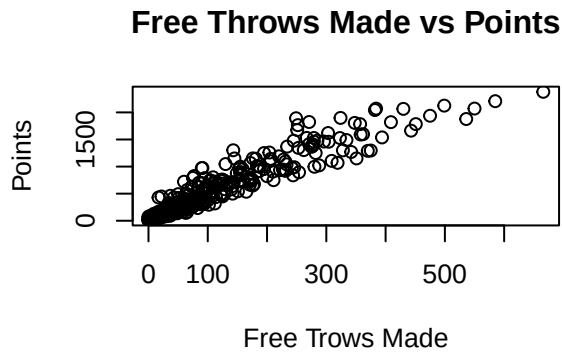
```



```

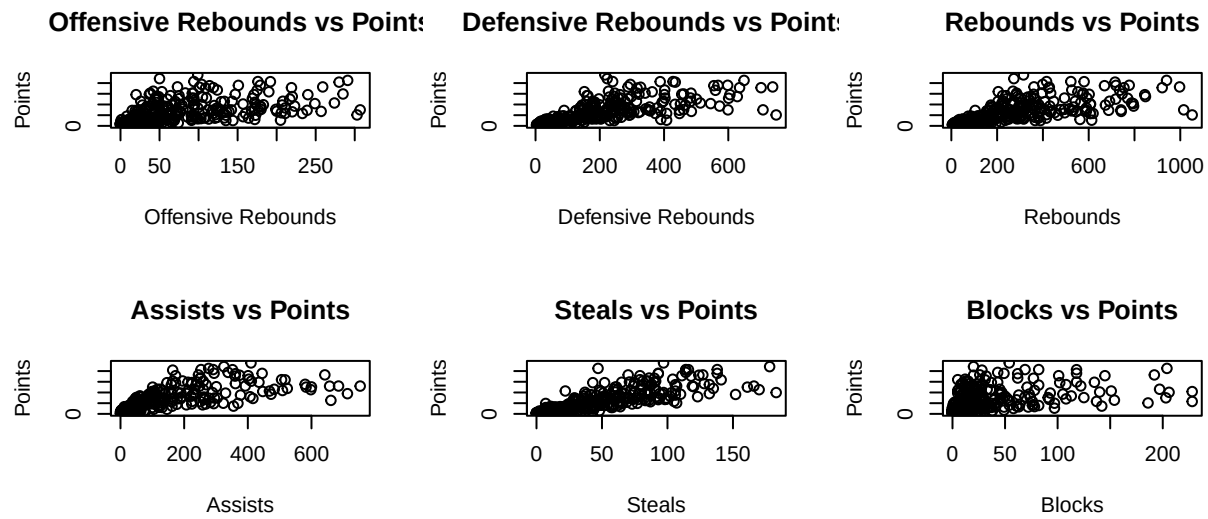
par(mfrow=c(2,2))
plot(x=player_sub$FTM, y=player_sub$PTS,
xlab="Free Throws Made", ylab="Points", main="Free Throws Made vs Points")
plot(x=player_sub$FTA, y=player_sub$PTS,
xlab="Free Throw Attempt", ylab="Points",main="Free Throw Attempt vs Points")
plot(x=player_sub$TOV, y=player_sub$PTS,
xlab="Turnovers", ylab="Points", main="Turnovers vs Points")
plot(x=player_sub$PF, y=player_sub$PTS,
xlab="Personal Fouls", ylab="Points", main="Personal Fouls vs Points")

```



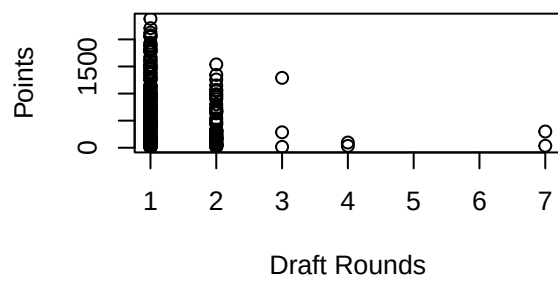
```
par(mfrow=c(3,3))
plot(x=player_sub$ORB, y=player_sub$PTS,
     xlab="Offensive Rebounds", ylab="Points", main="Offensive Rebounds vs Points")
plot(x=player_sub$DRB, y=player_sub$PTS,
     xlab="Defensive Rebounds", ylab="Points", main="Defensive Rebounds vs Points")
plot(x=player_sub$REB, y=player_sub$PTS,
     xlab="Rebounds", ylab="Points", main="Rebounds vs Points")
plot(x=player_sub$AST, y=player_sub$PTS,
     xlab="Assists", ylab="Points", main="Assists vs Points")
plot(x=player_sub$STL, y=player_sub$PTS,
     xlab="Steals", ylab="Points", main="Steals vs Points")
plot(x=player_sub$BLK, y=player_sub$PTS,
     xlab="Blocks", ylab="Points", main="Blocks vs Points")

par(mfrow=c(2,2))
```

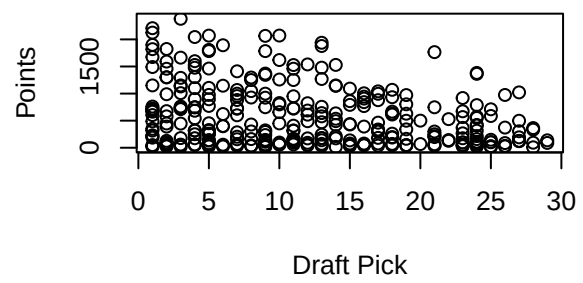


```
plot(x=player_sub$draft_round, y=player_sub$PTS,
     xlab="Draft Rounds", ylab="Points", main="Draft Rounds vs Points")
plot(x=player_sub$draft_pick, y=player_sub$PTS,
     xlab="Draft Pick", ylab="Points", main="Draft Pick vs Points")
```

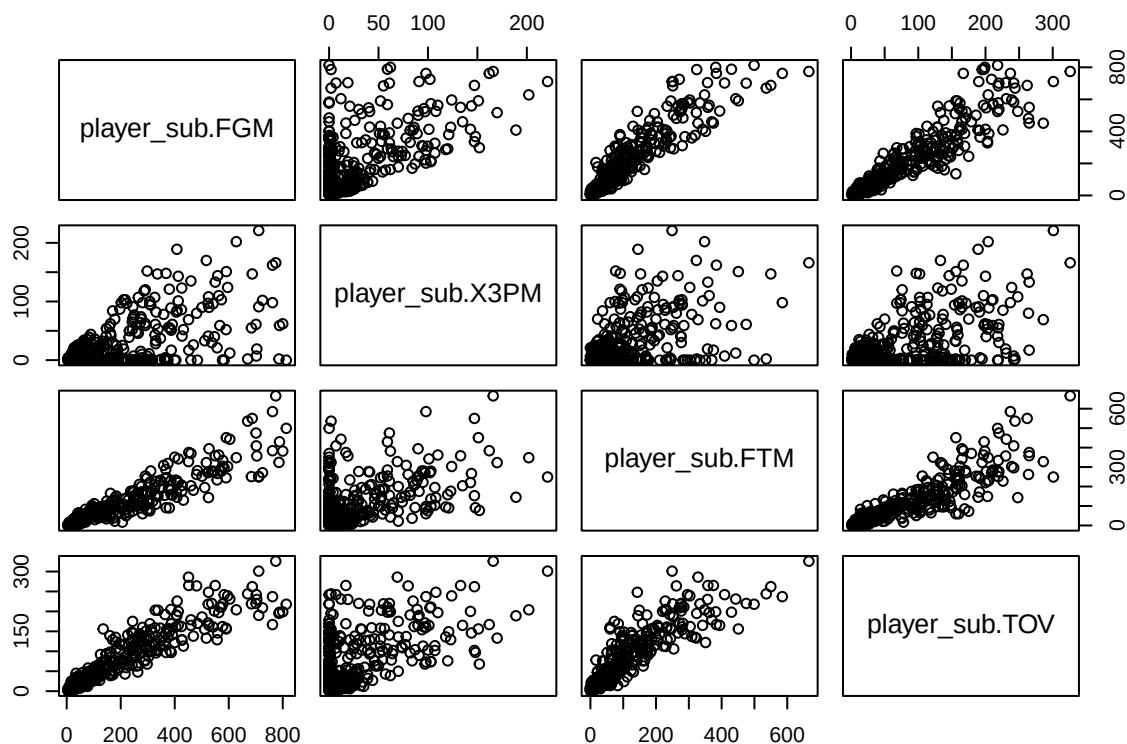
Draft Rounds vs Points



Draft Pick vs Points



```
columns=data.frame(player_sub$FGM, player_sub$X3PM, player_sub$FTM, player_sub$TOV)
pairs(columns)
```



Based on the exploratory plots shown above, the dependent variable with the strongest correlation are the following: Stage, field goals made and attempted, 3 points made, free throws made, turnovers, and draft rounds.

Model Building

The histogram of points (PTS) did not seem to be normally distributed, a good use of the model function to use is the logistic regression model (glm). While building the model, the first family used was Poisson distribution, since the default distribution of Gaussian is for when the dependent variable y is continuous and roughly normally distributed, which in this case is not, so a Poisson distribution was attempted.

After trying two different models of Poisson family, the dispersion test was done to determine if a Poisson model or a negative binomial model is a better fit for the data. Based on the overdispersion test, the dispersion ratio is 38.32936 and 71.05908 which is much greater than 1, so a negative binomial is a better fit for this data. The p-values were 5.634e-10 and <2.2e-16, which shows that these p-values $< \alpha(0.05)$, meaning there is significance of overdispersion.

As a result, the Poisson family is not appropriate for such model, so the negative binomial model was used and other models were attempted.

```
#poisson models
model0 <- glm(PTS~Stage+FGM+FGA+X3PM+FTM+TOV+draft_round,
              data=player_sub, family="poisson")
dispersiontest(model0, alternative="greater") #dispersion test
```

```
##
## Overdispersion test
##
## data: model0
## z = 6.0903, p-value = 5.634e-10
## alternative hypothesis: true dispersion is greater than 1
## sample estimates:
## dispersion
## 38.32936

model00 <- glm(PTS~FGM+X3PM+FTM+AST, data=player_sub, family="poisson")
dispersiontest(model00, alternative="greater") #dispersion test
```

```
##
## Overdispersion test
##
## data: model00
## z = 20.682, p-value < 2.2e-16
## alternative hypothesis: true dispersion is greater than 1
## sample estimates:
## dispersion
## 71.05908
```

```
#negative binomial models
model1 <- glm.nb(PTS~Stage+FGM+FGA+X3PM+FTM+TOV+draft_round,
                 data=player_sub)

model2 <- glm.nb(PTS~MIN+FGM+PF+ORB+DRB+AST+STL+BLK+draft_pick,
                 data=player_sub)

model3 <- glm.nb(PTS~FGM+X3PM+FTM+AST+draft_round, data=player_sub)

model4 <- glm.nb(PTS~FGM+X3PM+FTM+draft_round+draft_pick, data=player_sub)
```

Model Evaluation

From the AIC and BIC shown below, the model with the lowest AIC and BIC is model 1. Model 1 has an AIC of 4,068.454 and a BIC is 4,106.137, compared to model 4, which has one of the highest BIC found at 4236.585, which means model 1 is considered as the best fit for the data set “player_sub_by_season_full_details.csv”.

```
AIC(model1, model2, model3, model4)
```

```
##      df      AIC
## model1 10 4068.454
## model2 11 4070.055
## model3  7 4210.069
## model4  7 4210.207
```

From the AIC table above, the lowest AIC model is model 1, and the highest is model 4.

```
BIC(model1, model2, model3, model4)
```

```
##          df          BIC
## model1 10 4106.137
## model2 11 4111.507
## model3  7 4236.448
## model4  7 4236.585
```

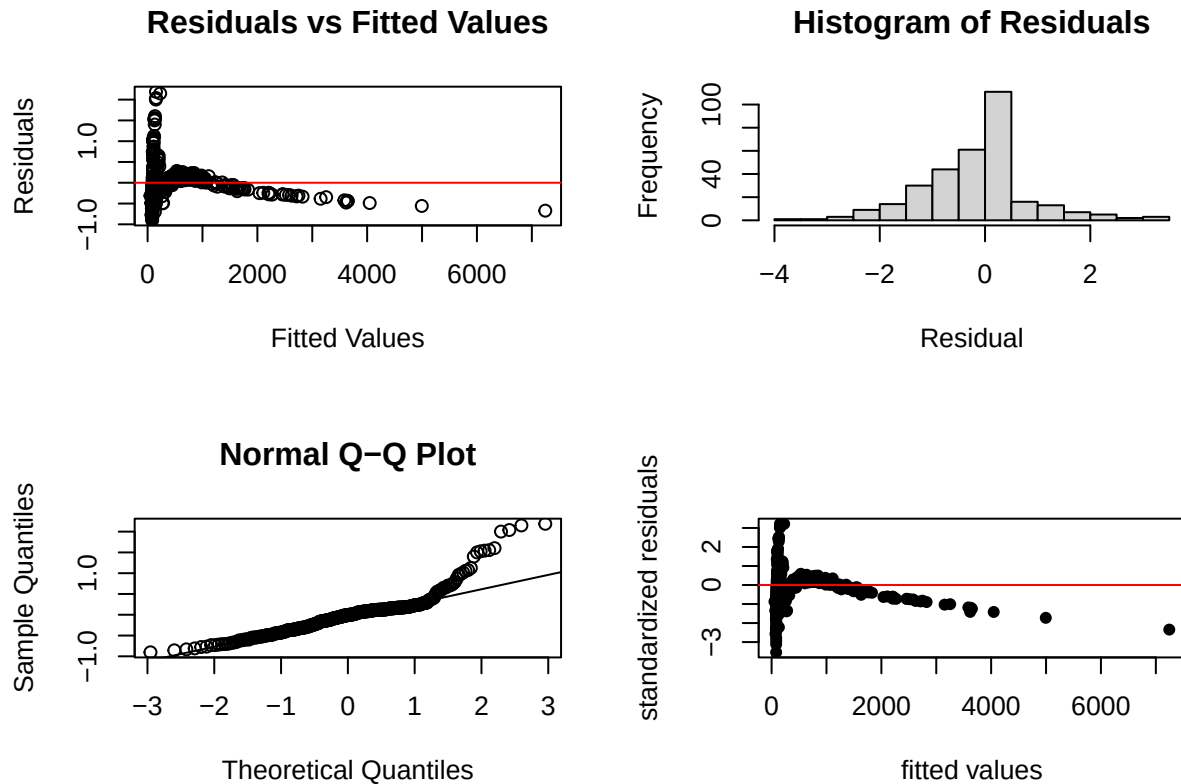
From the BIC table above, the lowest BIC model is model 1, and the highest is model 4.

Constant variance, normally distributed residuals, and linear relationships:

As seen on the plots below, the histogram of residuals are independent, identical normally distributed, there is also constant variance. The QQ plot points are mostly along the line, which shows that there is no systematic trend, and are centered around zero with no large outliers and are approximately normally distributed, showing a linear trend between predictors and response.

Based on the Exploratory Plots from above, the plots with the strongest linear relationships are as follows: Stage, field goals made and attempted, 3 points made, free throws made, turnovers, and draft rounds.

```
par(mfrow=c(2,2))
plot(model1$fitted.values, model1$residuals,
     main = "Residuals vs Fitted Values",
     xlab = "Fitted Values",
     ylab = "Residuals")
abline(h = 0, col = "red", lwd=1)
hist(resid(model1), xlab="Residual", main="Histogram of Residuals")
qqnorm(model1$residuals); qqline(model1$residuals)
plot(fitted(model1), rstandard(model1), xlab="fitted values", ylab="standardized residuals", pch=16)
abline(h=0, col="red", lwd=1)
```



Analysis and Findings

The data set “player_sub_by_season_full_details.csv”, shows the information of each season for different leagues and teams, with detailed information of the players and the games. In this analysis, the dependent outcome variable points made by the independent variables chosen are fitted into models that best describe the data set above.

After exploring the different plots, the distribution of dependent variable points was non-normal, this plot shows that the linear regression (lm) would not be a good model to use, such why the logistic regression (glm) would be a better model. The debate of using poisson or negative binomial was taken place as both are used for count (positive) and non-normally distributed data, as for the binomial family is solitary for binary (0 or 1) variables. For which in this case, a model0 and model00 of poisson family was created to test the overdispersion, which is if the variance is greater than the mean. Following this, the overdispersion ratio is 38.32936 and 71.05908, which shows that there is the model is overdispersed, and a negative binomial model should be used.

The following models 1-4 were taken with different dependent variables used, showing the different results for each model. In the model evaluation, the lowest AIC and BIC model was model 1 at a AIC of 4068.454 and a BIC is 4106.137. The highest model was model 4, with AIC of 4210.207 and BIC of 4236.585.

The residuals, fitted plots and QQ plots were also taken into consideration for model 1 with the lowest AIC and BIC to examine if the model would be normally distributed. In which case, the histogram of residuals were independent, identical and normally distributed with constant variance. The QQ plots were also normal, with a large amount of the points along the plotted line. Comparing the different plots in the

Exploratory Analysis, there is a strong linear relationship between points and all the different predictors that were added to the model 1.

From the finding in the analysis done above, the independent variables that the dependent variable points are dependent on are stage, field goals made, field goals attempted, 3 points made, free throws made, turnovers and draft rounds (Stage+FGM+FGA+X3PM+FTM+TOV+draft_round).

In conclusion, model 1 which has all the dependent variables mentioned above has the best model to analyse the dependent variable points.